



ELECT MAINT- OFF SITE Department
SIMHADRI



Ref:SMTTP/ELECT MAINT-/2022-23/DC SYSTE/270219

Date:

12/07/2022(18:54:43)

Subject:Revision of LMI/OIN/OPS/ELEC/046:MAINTENANCE & MANAGEMENT OF DC SYSTEM

Sub:-Revision of LMI/OIN/OPS/ELEC/046: MAINTENANCE & MANAGEMENT OF DC SYSTEM

The operation information note on Maintenance & Management of DC System has been revised vide COS-ISO-00-OIN/OPS/ELEC/046 Rev. No.: 1, June 2022.

Accordingly LMI/OIN/OPS/ELEC/046 has been revised with the inclusion of following revision details.

- 1) Thermoscanning of battery cell interconnections and cable connections to be done at regular interval during boost charging and discharging of battery banks during capacity test.
- 2) Thermoscanning of charger's power circuit connections to be done during boost charging of battery banks.
- 3) Scrap battery cells and electrolyte to be disposed safely in accordance with Battery Waste Management rules.

Head of the project is the competent authority for approval of this document.

Put up for approval please.

Submitted by:

जे एस एन वी प्रसाद - वरिष्ठ प्रबंधक

JULUVA SATYANARAYANA VARA PRASAD - 068631 (SENIOR MANAGER ,ELECT MAINT- OFF SITE) ,Simhadri

(JSNVPRASAD@NTPC.CO.IN, 9428815820)

On:12/07/2022(18:54:43) **Ref:**SMTPP/ELECT MAINT-/2022-23/DC SYSTE/270219

Modified LMI reviewed

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:13/07/2022(17:47:02) **Ref:**SMTPP/ELECT MAINT-/2022-23/DC SYSTE/270219

Reviewed

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:13/07/2022(18:03:13) **Ref:**SMTPP/ELECT MAINT-/2022-23/DC SYSTE/270219

Submitted for approval of competent authority

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
(JHIMLIDHAR@NTPC.CO.IN, 9650998085) ,Simhadri

On:13/07/2022(19:37:40) **Ref:**SMTTP/ELECT MAINT-/2022-23/DC SYSTE/270219

May please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:17/07/2022(08:32:06) **Ref:**SMTTP/ELECT MAINT-/2022-23/DC SYSTE/270219

Submitted by:

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,OPERATION)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:18/07/2022(12:32:53) **Ref:**SMTTP/ELECT MAINT-/2022-23/DC SYSTE/270219

समीक्षित। Reviewed.

कृपया अनुमोदन प्रदान किया जाए। May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
(RKJHA02@NTPC.CO.IN, 9415342046) ,Simhadri

On:18/07/2022(14:32:17) **Ref:**SMTTP/ELECT MAINT-/2022-23/DC SYSTE/270219

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

Girish Chandra Choukse - 002907 (GENERAL MANAGER ,HEAD OF PROJECT)
(GCCHOUKSE@NTPC.CO.IN, 9425178006) ,Simhadri

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NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LMI/OIN/OPS/ELEC/046
ISSUE NO: 1, REVISION: 2
DATE: JULY 2022

**MAINTENANCE & MANAGEMENT
OF
DC SYSTEM**

APPROVED FOR
IMPLEMENTATION BY - - - - -
HEAD OF THE PROJECT (SIMHADRI)

Date: _____

NTPC Limited
SIMHADRI SUPER THERMAL POWER STATION

OPERATION & MAINTENANCE DEPARTMENT

LMI (Name) : MAINTENANCE & MANAGEMENT OF DC SYSTEM			
LMI No. : LMI /OIN/OPS/ELEC/046			
Issue No.	1	Revision: 2	Date: JULY 2022

	Name	Designation	Signature
Prepared by	J S N V PRASAD	SR. MANAGER(EM)	
Reviewed by	T VIJAYA KRISHNA	AGM- (EM)	
Reviewed by	D K TOMAR	HOD- (EM)	
Reviewed by	MATHEW EIPE KOVOOR	GM (MAINTENANCE)	
Reviewed by	D P PATRA	GM(OPERATION)	
Reviewed by	R K JHA	GM (O&M)	

MAINTENANCE & MANAGEMENT OF DC SYSTEM

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1.0 OBJECTIVE :

DC System is the back bone of emergency supply equipment for power plant. Proper maintenance & management of DC System will not only ensure the healthiness of the system but also guarantees full capability & reliability of the DC system in case of emergency. The system in service should remain capable of discharging its full energy for the specified period at any point of time whenever called for and will not jeopardize safe shutdown of units.

The objective of this LMI is to specify various maintenance activities and their schedules and the management required to keep the DC System in “ Ready to Perform with full capacity” condition.

2.0 SCOPE :

This document covers various activities to be carried out during service / opportunity shut down to keep the DC supply system in its healthiest condition to cater any exigencies. This LMI covers the Ni-Cd battery banks of various makes / types used for providing DC power supplies at NTPC Simhadri.

3.0 REFERENCE DOCUMENTS

COS-ISO-00-OIN/OPS/ELEC/046 Rev. No.: 1, June 2022

4.0 SUPERCEDED DOCUMENTS :

OIN-OPS-ELEC-046 Rev 0 Dated: September 2009

5.0 SPECIAL INSTRUCTIONS TO BE FOLLOWED

Instruction	Responsibility	Document Ref.
➤ Daily Monitoring of DCDB, Charger and Battery Voltages	Operation TG desk engineer / EMD Area Engineer	OIN-OPS-ELEC-046
➤ Monthly monitoring of DC System battery voltages		
➤ Monitoring and maintenance of DC System during overhaul		

6.0 REVISION DETAILS

Instruction	Responsibility
➤ Thermoscanning of battery cell interconnections and cable connections to be done at regular interval during boost charging and discharging of battery banks during capacity test ➤ Thermoscanning of charger's power circuit connections to be done during boost charging of battery banks. ➤ Scrap battery cells and electrolyte to be disposed safely in accordance with Battery Waste Management rules.	Area Engineer / Section head

7.0 METHODOLOGY OF IMPLEMENTATION

(a) Daily Monitoring & Management of DC System

Equipment	Activity	Responsibility	Record	Corrective action
DCDB	Checking of DCDB P-E, N-E & P-N voltages for DC earth fault	EMD Area Engineer	P-E, N-E, P-N	Operation TG desk engineer must inform to Area engineer on appearance of DC earth fault alarm. Immediately attend the DC earth fault
DC Charger	Trickle / Float charging current @2.0 mA/ Ah	EMD Area Engineer	O/P current	Adjust float voltage if required
Battery	Checking of Pilot cell voltage @ 1.40 to 1.42 V/ cell	EMD Area Engineer	P-N	- Adjust float voltage if required. - Equalizing charge if required
DC Drives	Trial run of DC drives	Operation TG desk engineer	OK/ NOT OK	Inform concerned mtc engineer if any abnormality is observed.

(b) Monthly monitoring & Management of DC System

Equipment	Activity	Responsibility	Record	Corrective action
Battery	Checking of electrolyte level and any spillage / crack cells	EMD Area Engineer	OK/ NOT OK	1. Replace cracked cell 2. Top up DM water if required
	General appearance and cleanliness of the battery and battery area	Operation TG desk engineer / EMD Area Engineer	OK/ NOT OK	-When excessive dirt deposition is noted, wipe with clean water. -Identify the source cause of ingress & rectify the same
	Any evidence of sulphation / corrosion at either terminals or connectors	EMD Area Engineer	Cell no.	Remove sulphation and apply petroleum jelly.
Battery Room	Condition of ventilation equipment	Operation TG desk engineer	OK/ NOT OK	Rectify problem immediately
	Availability of tap water / eye wash safety shower in battery room	EMD Area Engineer	OK/ NOT OK	Rectify problem immediately

Equipment	Activity	Responsibility	Record	Corrective action
DC Charger	Charger output current and voltage	EMD Area Engineer	O/P Voltage O/P Current	Adjust float voltage set point if required
Battery	Voltage of Each cell and total battery terminal voltage	EMD Area Engineer	P-N	- Adjust float voltage if required. -Equalizing charge if required

(c) Half Yearly (during opportunity shut down) monitoring &
Management of DC System

Equipment	Activity	Responsibility	Record	Corrective action
Battery	Boost charging of battery sets	EMD Area Engineer	Cell no. Voltage.	
DCDB	Checking for AC/DC mixing in main DCDBs of units and Switchyard	EMD Area Engineer	OK/ NOT OK	Immediately attend to rectify the abnormality.

(d) Biennial (during respective unit overhaul) monitoring &
Management of DC System

Equipment	Activity	Responsibility	Record	Corrective action
Battery	Capacity test	EMD Area Engineer	Capacity	
	Tightness of bolted connections to recommended torque value followed by cell to cell and terminal connection detail resistance	EMD Area Engineer	Resistance (Cell-Cell) (Terminal connection)	
	Inspect integrity of battery racks, cell condition etc.	EMD Area Engineer	Defects	
Chargers	Load test of chargers & overhauling	EMD Area Engineer	OH report	
	Ripple content measurement of chargers (3-4%)	EMD Area Engineer	Ripple %	Record and compare with charger specs. If ripple is high, check & rectify filters.
DCDB	Overhauling of DCDB	EMD Area Engineer	OH report	
DC feeders	Checking for DC-DC mixing	EMD Area Engineer	OK/ NOT OK	During annual shut down, one source to be made off in panels having two sources and

Equipment	Activity	Responsibility	Record	Corrective action
				voltage (P-E, N-E) in the downstream circuit to be measured / checked for ant mixing.
DC Drives.	Trial run of DC drives and speed measurement	Operation TG desk engineer.	Speed of DC Drive	- Check starting resistors cut in / out. -Clean commutators -Replace carbon brushes - check cable tightness - check control & power circuits.
DC System	Load testing of DC System on running of all DC drives for at least one hour, keeping charger off & checking of healthiness.	EMD Area Engineer	Terminal Voltage of Battery set after one hr.	Equalizing charge if required. Thermoscanning of batteries and cable Connections during load testing to be carried out

8.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Sl. No	Activity	Responsibility
8.1	Implementation of LMI	Heads of EM and Operation
8.2	Carrying out modifications	Head of EM
8.3	Overall Co-ordination	Head of EM

9.0 EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS

Any exceptions / deviations shall be reported to HOD (EM), HOD (OPERATION).

10.0 RECORD KEEPING:

Records will be maintained by EMD area maintenance engineer.

11.0 REVIEW:

This LMI document will be reviewed once in every two years by Head of the Project, NTPC Simhadri.